

LOGICAL REASONING - ADVANCED MCQ PAPER

Mid-Term Examination | Total Marks: 25 | Duration: 1 Hour 30 Minutes
Each question carries 1 mark. No negative marking. Choose the best answer.
DIFFICULTY: MODERATE TO HARD | Based on actual exam-level problems

SECTION A: NUMBERS (3 marks)

Q1. The number $27942^*@$ is exactly divisible by 11. If * and @ are two different single-digit numbers, how many such numbers exist?

- a) 9
- b) 10
- c) 12
- d) 1

Q2. $N = 2^5 \times 3^2 \times 5^1$. Find the number of factors and the sum of all factors of N.

- a) 36 factors, Sum = 4914
- b) 30 factors, Sum = 4914
- c) 36 factors, Sum = 4824
- d) 30 factors, Sum = 4824

Q3. What is the remainder when $65347 \times 74789 \times 57743$ is divided by 4?

- a) 1
- b) 2
- c) 3
- d) 0

SECTION B: PERCENTAGE (3 marks)

Q4. Fresh grapes contain 90% water by weight. Dried grapes contain 20% water. What is the weight of dried grapes obtained from 20 kg of fresh grapes?

- a) 2 kg
- b) 2.4 kg
- c) 2.5 kg
- d) 3 kg

Q5. After two successive raises, Ahmed's salary becomes $\frac{75}{40}$ times the original. If the second raise is twice the first raise (in %), find the first raise.

- a) 20%
- b) 25%
- c) 30%
- d) 50%

Q6. The price of a commodity increases by 20%. By what percent should consumption be reduced so that total expenditure increases by only 8%?

- a) 8%
- b) 10%
- c) 12%
- d) 15%

SECTION C: PROFIT AND LOSS (3 marks)

Q7. An article is sold at 30% profit. If it was bought at 10% less and sold for Rs. 40 less, a 40% profit would be gained. Find the original cost price.

- a) Rs. 500

- b) Rs. 800
- c) Rs. 1000
- d) Rs. 1200

Q8. A shopkeeper claims to sell at 8% loss but uses a weight of 800 gm for 1 kg. His actual profit or loss percentage is:

- a) 8% loss
- b) 15% profit
- c) 10% profit
- d) 12% profit

Q9. A merchant sells at 30% profit. After allowing 10% discount, his sales improved from ratio 2 to 5. The ratio of original profit to new profit is:

- a) 12 : 17
- b) 17 : 12
- c) 13 : 12
- d) 12 : 13

SECTION D: AVERAGE (3 marks)

Q10. Ten years ago, a joint family of 8 had total age 231 years. Three years later, one member died at 60 and a child was born. Three years after that, another died at 60 and a child was born. The current average age is nearest to:

- a) 23 years
- b) 22 years
- c) 21 years
- d) 24 years

Q11. Amol takes the average of 10 two-digit numbers. By mistake he interchanges the digits of one number, and the average increases by 1.8. If the digits of that number are a and b (tens and units), find (b - a):

- a) 3
- b) 1
- c) 2
- d) 4

Q12. A CLAT aspirant's average score increases by 1 if first 10 tests are excluded, and decreases by 1 if last 10 are excluded. Average of first 10 tests = 20, last 10 = 30. Total number of tests:

- a) 40
- b) 50
- c) 60
- d) 80

SECTION E: PERMUTATION & COMBINATION (4 marks)

Q13. How many 5-digit numbers divisible by 4 can be formed from digits 1, 2, 3, 4, 5, 6 without repetition?

- a) 144
- b) 168
- c) 192
- d) 186

Q14. If the letters of MOTHER are arranged as in a dictionary, the rank of the word MOTHER is:

- a) 240
- b) 261
- c) 308

d) 309

Q15. Ten points lie on one straight line and eleven on another. How many triangles can be formed with vertices from these 21 points?

- a) 495
- b) 550
- c) 1045
- d) 2475

Q16. There are 6 tasks and 6 persons. Task 1 cannot be assigned to person 1 or 2. Task 2 must be assigned to person 3 or 4. Each person gets exactly one task. Number of ways:

- a) 144
- b) 180
- c) 192
- d) 360

SECTION F: PROBABILITY (4 marks)

Q17. The digits 1-9 are written in random order to form a 9-digit number. The probability it is divisible by 4 is:

- a) $\frac{4}{9}$
- b) $\frac{2}{9}$
- c) $\frac{1}{3}$
- d) $\frac{5}{18}$

Q18. In shuffling 52 cards, 4 are accidentally dropped. Probability all 4 are from different suits is:

- a) $\frac{2197}{20825}$
- b) $\frac{169}{1225}$
- c) $\frac{13}{52}$
- d) $\frac{2197}{52000}$

Q19. Odds are 11:5 against a 38-year-old living to 73, and 5:3 against a 43-year-old living to 78. The probability at least one survives 35 more years is:

- a) $\frac{1}{128}$
- b) $\frac{15}{128}$
- c) $\frac{73}{128}$
- d) $\frac{55}{128}$

Q20. To get at least one head with probability ≥ 0.9 , the minimum number of times a fair coin must be tossed is:

- a) 2
- b) 3
- c) 4
- d) 5

SECTION G: WORK & CHAIN RULE (2 marks)

Q21. A can do a piece of work in 12 days. B is 60% more efficient than A. The number of days B alone takes to finish the work is:

- a) 6 days
- b) 7 days
- c) 7.5 days
- d) 8 days

Q22. 15 men working 8 hours/day can complete a work in 20 days. How many days will 20 men working 6 hours/day take to complete the same work?

- a) 18 days
- b) 20 days
- c) 24 days
- d) 15 days

SECTION H: SERIES (2 marks)

Q23. Find the next number: 3, 8, 18, 35, 61, ?

- a) 88
- b) 95
- c) 98
- d) 100

Q24. In the series Z, W, T, Q, N, ?, which letter comes next?

- a) J
- b) K
- c) L
- d) M

SECTION I: CODING - DECODING (1 mark)

Q25. In a code language, TIGER is written as UJHFS. Using the same pattern, HORSE is written as:

- a) IPSFT
- b) IPSTF
- c) IPSTF
- d) IPTFS

ANSWER KEY WITH DETAILED EXPLANATIONS

Q.	Answer	Q.	Answer	Q.	Answer
Q1	a) 9	Q2	a) 36 factors, Sum = 4914	Q3	a) 1
Q4	c) 2.5 kg	Q5	b) 25%	Q6	b) 10%
Q7	c) Rs. 1000	Q8	b) 15% profit	Q9	a) 12 : 17
Q10	d) 24 years	Q11	c) 2	Q12	c) 60
Q13	c) 192	Q14	d) 309	Q15	c) 1045
Q16	a) 144	Q17	b) 2/9	Q18	a) 2197/20825
Q19	c) 73/128	Q20	c) 4	Q21	c) 7.5 days
Q22	b) 20 days	Q23	c) 98	Q24	b) K
Q25	c) IPSTF				

Step-by-Step Solutions:

Q1. Divisibility by 11: $(2+9+2+@) - (7+4+*) = 0$ or multiple of 11. So $13+@ = 11+* \Rightarrow @-* = -2$ OR $13+@ - 11-* = 11 \Rightarrow @-* = 9$. From $@-* = -2$ (i.e., $*=@+2$): pairs (0,2),(1,3),(2,4),(3,5),(4,6),(5,7),(6,8),(7,9) = 8 pairs. From $@-* = 9$: only (9,0) = 1 pair. Total = 9.

Q2. Factors = $(5+1)(2+1)(1+1) = 36$. Sum = $[(2^6 - 1)/(2 - 1)] \times [(3^3 - 1)/(3 - 1)] \times [(5^2 - 1)/(5 - 1)] = 63 \times 13 \times 6 = 4914$.

Q3. Only last 2 digits matter for div by 4. $47\%4=3$, $89\%4=1$, $43\%4=3$. Product of remainders = $3 \times 1 \times 3 = 9$. $9\%4 = 1$.

Q4. Pulp in 20 kg fresh = 10% of 20 = 2 kg. In dried grapes, pulp = 80%. So 80% of dried weight = 2 kg $\Rightarrow$ dried = $2/0.8 = 2.5$ kg.

Q5. Let first raise = R%. Second = 2R%. So $(1+R/100)(1+2R/100) = 75/40 = 15/8$. Expanding: $(100+R)(100+2R) = 187500/100$. Solving $2R^2 + 300R - 8750 = 0 \Rightarrow R^2 + 150R - 4375 = 0 \Rightarrow (R+175)(R-25) = 0 \Rightarrow R = 25\%$.

Q6. Let original price = P, consumption = C. New: $1.2P \times (1-x/100) \times C = 1.08 \times PC$. So $1.2(1-x/100) = 1.08$. $1-x/100 = 0.9$. $x = 10\%$.

Q7. Let CP = 100x. SP = 130x. New CP = 90x. New SP = $90x \times 1.4 = 126x$. Difference = $130x - 126x = 4x = 40$. So $x = 10$, CP = Rs. 1000.

Q8. Effective SP for 800 gm = 92% of 1000 gm price = Rs. 920 (for every Rs. 1000 of goods). But actual goods given = 800 gm worth Rs. 800. Profit = $(920-800)/800 \times 100 = 15\%$.

Q9. Let CP/item = 100. Original: SP = 130, sales = 2 items, profit = 60. New: SP = $130 \times 0.9 = 117$, sales = 5 items, profit = $17 \times 5 = 85$. Ratio = $60:85 = 12:17$.

Q10. 10 yrs ago: total = 231. After 3 yrs: $231+24-60+0 = 195$. After 3 more: $195+24-60+0 = 159$. Now (4 yrs later): $159+32 = 191$. Avg = $191/8 \sim 24$ years.

Q11. Increase in sum = $1.8 \times 10 = 18$. Interchanged - Original = $(10b+a) - (10a+b) = 9(b-a) = 18$. So $b-a = 2$.

Q12. Let N = total tests, A = overall average. $(NA-200)/(N-10) = A+1$ and $(NA-300)/(N-10) = A-1$. Subtracting: $100/0 \dots$ Actually: From eq1: $NA-200 = (A+1)(N-10)$; From eq2: $NA-300 = (A-1)(N-10)$. Subtracting: $100 = 2(N-10) \Rightarrow N = 60$.

Q13. Last 2 digits must form a number divisible by 4: 12,16,24,32,36,52,56,64 = 8 possibilities. Remaining 3 digits from 4 left = $4 \times 3 \times 2 = 24$. Total = $8 \times 24 = 192$.

Q14. Words starting with E,H: 120 each = 240. ME,MH: 24 each = 48. MOE,MOH,MOR: 6 each = 18. MOTEHR: 1, MOTERH: 1, MOTHER: 1. Total = $240+48+18+3 = 309$.

Q15. Pick 2 from line 1 and 1 from line 2: $10C2 \times 11C1 = 45 \times 11 = 495$. Or 1 from line 1 and 2 from line 2: $10C1 \times 11C2 = 10 \times 55 = 550$. Total = $495 + 550 = 1045$.

Q16. Task 2: 2 choices (person 3 or 4). Task 1: cannot go to person 1 or 2, and one of 3/4 is taken, so 3 choices. Remaining 4 tasks to 4 people = $4! = 24$. Total = $2 \times 3 \times 24 = 144$.

Q17. Number divisible by 4 if last 2 digits form number div by 4. From digits 1-9, pairs div by 4: 12,16,24,28,32,36,48,52,56,64,68,72,76,84,92,96 = 16 pairs. Each has 7! arrangements of remaining. Total arrangements = $9!$. $P = 16 \times 7! / 9! = 16/72 = 2/9$.

Q18. Favorable: $13C1 \times 13C1 \times 13C1 \times 13C1 = 13^4 = 28561$. Total: $52C4 = 270725$. $P = 28561/270725 = 2197/20825$.

Q19. $P(A \text{ survives}) = 5/16$, $P(B \text{ survives}) = 3/8$. $P(\text{at least one}) = 1 - P(\text{none}) = 1 - (11/16)(5/8) = 1 - 55/128 = 73/128$.

Q20. $P(\text{at least 1 head}) = 1 - (1/2)^n \geq 0.9$. So $(1/2)^n \leq 0.1$. $n=3$: $1/8=0.125 > 0.1$ (fails). $n=4$: $1/16=0.0625 \leq 0.1$ (works). Min $n = 4$.

Q21. B is 60% more efficient $\Rightarrow$ B's rate = $1.6 \times$ A's rate. A's time = 12 days. B's time = $12/1.6 = 7.5$ days.

Q22. $M1 \times D1 \times H1 = M2 \times D2 \times H2$. $15 \times 20 \times 8 = 20 \times D2 \times 6$. $D2 = 2400/120 = 20$ days.

Q23. Differences: 5, 10, 17, 26. Second differences: 5, 7, 9 (AP, diff=2). Next 2nd diff = 11. Next 1st diff = $26+11 = 37$. Answer = $61+37 = 98$.

Q24. Z(26), W(23), T(20), Q(17), N(14), ?(11). Each letter is -3 positions. $14-3 = 11 = K$.

Q25. Each letter shifted +1: T $\rightarrow$ U, I $\rightarrow$ J, G $\rightarrow$ H, E $\rightarrow$ F, R $\rightarrow$ S. So H $\rightarrow$ I, O $\rightarrow$ P, R $\rightarrow$ S, S $\rightarrow$ T, E $\rightarrow$ F = IPSTF.